

# Faculty of Technology, Design and Environment

## MASTER OF SCIENCE DISSERTATION

|                      |                                                          |
|----------------------|----------------------------------------------------------|
| Dissertation Title:  | Sports Analysis: Football Player Market Value Prediction |
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| Dissertation Module: | COMP7039                                                 |
| Course Title:        | Dissertation in Computing Subject                        |
| Date Submitted:      | 26/September/2025                                        |

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Except for those parts in which it is explicitly stated to the contrary, this project is my own work. It has not been submitted for any degree at this or any other academic or professional institution.

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## 1. Glossary

Gradient Boosting Regression: ensemble predictive model.

Linear Regression

Lasso Regression

Ridge Regression

Random Forest

Feature Importance: indicates relative influence on prediction.

Squad: Club a player represents

MAE: Mean Absolute error

MSE: Mean Squared error

RMSE: Root Mean Squared error

$R^2$ : R-Squared

Age<sup>2</sup> (Age Squared): player's age capturing non-linear effects.

Ast (Assists): Final pass to teammate scoring a goal.

xG: Expected Goals

xAG: Expected Assits

npxG: Non-penalty expected goals

npxG + xAG.

Gls: Goals

PrgP: Progressive passes

Nation\_Eng: Categorical feature for English Nationality.

MP: Matches Played

Min: minutes played

## 2. Abstract

The evaluation of football player market worth has developed as the primary emphasis in both academic research and professional practice, because of the current transfer market's significant financial stakes. Conventional methodologies, which are often based on subjective scouting judgments or crowd-sourced appraisals, sometimes fail to reflect underlying performance measures and complicated market dynamics. This dissertation presents a data-centric method for player valuation that uses machine learning techniques, namely ensemble modeling, to improve forecast accuracy and interpretability.

A large dataset of player attributes, performance measurements, and contextual factors is used to evaluate many prediction models, including Linear Regression, Ridge, Lasso, Random Forest, and Gradient Boosting Regression. Gradient Boosting outperformed linear approaches in terms of prediction accuracy ( $R^2 = 0.70$ ). In addition to its predictive power, the model also provided useful interpretability by showing the relative importance of key variables on player market valuation.

The study demonstrated that progressive passes, age profile (age<sup>2</sup>), non-penalty anticipated goals plus assisted goals (npAG + xAG), nationality, and minutes played were among the most important market value drivers. These findings highlight the rising relevance of sophisticated performance indicators and contextual considerations in setting values, which go beyond simple measures like goals and assists. The findings emphasize the non-linear character of player careers, the commercial and regulatory implications of nationality, and the importance of sustained underlying performance.

This study offers a practical implementation by combining contemporary football analytics and machine learning to provide a clear and interpretable value model. It has practical result for clubs, scouts, and executives seeking more transparent, data-driven approaches to navigate the complex nature of transfer market.

### **3. Acknowledgements**

I want to express my deep thanks to Mathias Rolf, my supervisor, for all of their help, advice, and constructive criticism during this study. Their knowledge was crucial in forming this work's ethical and technological aspects. I also want to thank my assessor, Ian Bayley, for his constructive feedback.

For providing the academic framework and resources that made this study possible, I am also appreciative of Oxford Brookes University. I would especially want to thank my family and friends for their patience and believe in me during this journey, as well as my classmates for their enlightening conversations and support.

## 4. Introduction

### 4.1. Background

Data Analytics has become an integral aspect of modern sports, it influences both on-field strategies and off-field business decisions. Football clubs invest heavily in talent acquisition, player development and transfer negotiations, making accurate assessment of player's market value in regards to maintaining competitive and financial advantage. Modern sports analysis uses digital tools like GPS trackers, wearable technology and high-speed cameras to get real-time, detailed data. In the past, sports analysis relied on simple scorecards, written observations and the coach's gut feeling. (Pliability, 2022) (Arastey, 2020).

Football player market value has been assessed through expert judgment, player reputation, performance metrics (appearance, goals, assists), age, contract length and a club's financial context. Generally, actual player transfer fees and market value estimates are based on comparison with past transfers, expert opinions and crowd-sourced data especially from platforms like Transfermarket. The procedure mainly depends on the subjective evaluations of scouts, managers, agents and crowdsourced estimates, which are frequently backed up by simple statistics and news reports (Arastey, 2020). Transfermarket and other sites like it use the opinions of both experts and regular people to figure out how much things are worth on the market. These values are usually very close to the actual transfer prices, but they can still be affected by things like negotiations, market trends and other outside factors e.g. international status. (Müller, Simons, & Weinmann, 2017).

With the growing databases like FBref and Transfermarket, modern research takes the advantage of ML models like Linear Regression, Gradient boosting and Multi-level regression to offer greater predictive power and transparency. Linear Regression models provide interpretability and expose direct correlations between features like age, position, performance with market values (Kong, 2025). Gradient Boosting has shown to achieve the highest accuracy and lowest RMSE among regression models, identifying intricate variable influences and offering robust predictions for unseen players. (Koen, Arem, Goes-Smit, & Söhl, 2025).

Analyzing player market value in football is a sophisticated and continually developing field, with major implications for clubs, agents and the wider football industry. It plays a key role in financial planning, team strategy and talent management (Felices, 2025). Also the growth of the transfer market was significantly boosted after the 1995 Bosman ruling. This shift prompted clubs to view players as key financial assets, integrals to their economic strategies and overall valuation. As a result, transfer fees surged worldwide and the approach to valuing players became more systematic, incorporating principles from economics, performance data and strategic planning (Statathlon).

Recent reviews of academic research have identified and grouped the main factors that influence a football players market value. These include time-related aspects (like age and contract length), labor contributions (such as minutes played and goals scored), popularity (media visibility), performance data (advanced match statistics), club-related features and personal player traits. Studies consistently show that factors like age typically peaking in value during prime years in match appearances, international experience and detailed performance metrics are strongly linked to market value. Meanwhile, less influential factors like disciplinary history or physical characteristics are being phased out of more reliable valuation models (Rong, Wang, & Xie, 2024) (Rivera, Santana, Burgueño, & Briseño., 2025).

Assessing the market value of football players is a complex task shaped by the changing dynamics of global transfer scene. Lack of comprehensive and reliable data such as contract specifics, agent commissions and the nuances of transfer negotiations are frequently unavailable or undisclosed, leading to distorted valuation models. Additionally, many performance metrics tend to overlap or correlate strongly, making it challenging for conventional statistical approaches to pinpoint the most influential variables with precision (Koen, Arem, Goes-Smit, & Söhl, 2025).

To address these challenges, this project utilizes advanced data techniques, machine learning algorithms and layered analytical approaches to enhance the accuracy of player market value predictions. It also uncovers the underlying factors that influence these valuations. By shifting away from basic metrics and subjective assessments, it provides football clubs with a more transparent, evidence-based view of how player values are determined in today's game.

#### **4.2. Aim**

The aim of this dissertation is to identify the features that influence players market value using the best performing prediction model, it focuses on designing and assessing various machine learning approaches to effectively estimate the market value of football players using a rich dataset sourced from platforms like Transfermarkt and FBref. It draws on a rich dataset that includes performance metrics, demographic details and contextual factors. The research addresses longstanding issues such as missing data, variable interdependence and the intricacies of the transfer market by utilizing advanced regression methods and ensemble learning techniques.

#### **4.3. Objectives**

Following are the objectives for this project:

- Collect and process the data of football players, combining performance statistics, demographic details and contextual club information relevant for market value estimation.
- Investigate, visualize and summarize how player market values are distributed and how they relate to various influencing factors, highlighting key patterns and identifying any anomalies.
- Construct a diverse set of machine learning models including linear regression, Ridge, Lasso, Random Forest, Gradient Boosting, XGBoost and hierarchical regression to estimate player market value, integrating both conventional and cutting-edge techniques.
- Assess each model's performance using established evaluation metrics such as RMSE and  $R^2$  to determine predictive accuracy and reliability, ultimately identifying the most effective modeling strategy.
- Identify the features that contribute to the model's performance, indicating the influence on the football player's market value.

#### **4.4. Product Overview**

This project is all about building a smart, data-driven system to estimate how much football players are worth. It covers everything from gathering and cleaning the data, to exploring patterns and making predictions and finally understanding what factors matter most in deciding a player's value. It implements a combination of fundamental and advanced machine learning techniques, examining both conventional and current methods of rating players. The purpose is not only to estimate prices, but also to help teams and analysts understand why players are valued as they are. Although it is focused on the English Premier League, the same methodology can be used to other leagues or sports.

This project is for a wide range of professionals in the football world, including those responsible for assessing players, handling transfers, managing budgets and making strategic decisions. Club managers and coaches get a data-driven way to evaluate player market values, moving beyond gut feelings and personal judgments. By combining in-depth performance data with relevant context, it supports smarter choices around player transfers, contract decisions and building the squad, ultimately helping teams perform better while staying within budget.

For recruitment analysts and scouts, the system pinpoints the key traits that most impact a player's market value, based on advanced machine learning analysis. This helps make scouting and talent evaluation more precise and data-driven, lowering the risks of poor signings and giving clubs a strategic edge.

For Financial and executive decision makers, this project helps assess the value of football players, making it easier to manage budgets, report finances and follow rules like financial fair play. By reducing guesswork around transfer costs and how player value changes over time, it supports smarter investment planning and better risk management.

In short, this product brings together analytics, explanations and intuitive visuals to tackle key issues in valuing football players. It helps everyone involved from coaches to executives make smarter, data-informed choices that boost competitiveness, support long-term financial health and improve strategic planning in the fast-paced world of football transfers.

## **5. Background Review**

As professional football continues to grow economically and more detailed player data becomes available, the topic of player valuation has gained significant interest in research. Recent work highlights the use of performance metrics, financial indicators and machine learning techniques to improve the precision and clarity of market value predictions.

(Franceschi, Brocard, Follert, & Gouguet, 2023) conducted a thorough systematic review of the key factors influencing football player valuation. Their analysis identified strong predictors such as age, playing time, match appearances, goals, assists and involvement with national teams. In order to increase prediction accuracy, the review also highlights the significance of club-level elements like financial resources and club prestige. Additionally, this research notes that many studies lack a strong theoretical basis, indicating that in order to gain a deeper understanding of the market, future research should integrate frameworks such as agency theory and human capital theory.

(Sun & Gu, 2024) investigated the valuation of football players across European leagues by utilizing a hybrid dataset that merged FIFA 19 video game attributes with actual performance data. Their findings revealed that integrating subjective ratings from FIFA with objective metrics significantly improves the accuracy of valuation models, striking a balance between data depth and realism. Similarly, (Al-Asadi & Tasdemir, 2022) employed machine learning methods on FIFA-based data and found that ensemble techniques such as Random Forest outperformed traditional linear models, underscoring machine learning's strength in capturing nonlinear patterns and complex interactions among features.

(Kong, 2025) carried out a comparative analysis of various machine learning models, employing thorough cross-validation techniques to predict football player market values. The study demonstrated that tree-based algorithms such as XGBoost and Gradient Boosting deliver superior performance, particularly in managing multicollinearity and capturing nonlinear relationships among player features. (Koen, Arem, Goes-Smit, & Söhl, 2025) conducted a longitudinal study to forecast player development and transfer values over time,

emphasizing the need for uncertainty estimation and model interpretability to support real-world decision-making within football clubs.

(Perez-Gonzalez, Fernandez-Luna, Castillo, & Burillo, 2020) examined the impact of the “relative age effect” (RAE) on the market valuation of European football players, a theory suggesting that athletes born earlier in the selection year may benefit from developmental advantages. Their findings showed that players born in the first quarter of the year generally receive higher market valuations than those born later, with this trend being most pronounced among top-tier athletes. This highlights how demographic and developmental factors, in addition to athletic performance, play a significant role in shaping player market value, reinforcing its complex and multifaceted nature.

(Nsolo, 2018) thesis presents an extensive review of predictive modeling techniques in soccer analytics, covering established methods like regression and time-series analysis, as well as newer machine learning approaches. The study underscores a shift from basic, easily interpretable models to more sophisticated algorithms capable of handling large, diverse datasets and uncovering hidden patterns that traditional statistical tools may overlook. Nsolo supports the adoption of ensemble learning and cross-validation strategies to enhance model precision and reliability.

(Müller, Simons, & Weinmann, 2017) questioned the dependence on crowd-sourced opinions and expert assessments in determining football players market value. Their study showed that data-driven approaches can yield estimates that closely match real transfer fees. This work supports the shift toward transparent, objective and reproducible analytics methods that can complement or even improve upon traditional valuation techniques.

The reliability and precision of predictive models depend heavily on strong evaluation metrics. (Olumide, 2023) emphasized the role of Root Mean Squared Error (RMSE) in measuring prediction accuracy, noting that lower RMSE values reflect a better-fitting model. Furthermore, (Koen, Arem, Goes-Smit, & Söhl, 2025) highlighted the growing importance of model explainability and uncertainty estimation, which are essential for building trust and delivering practical insights to football clubs. (Arastey, 2020), (Pliability, 2022), (Kong, 2025) also highlight how crucial model explainability and performance analysis are to converting forecasted outcomes into actual football decisions. Their work demonstrates how these components help close the gap between data outputs and strategic action by making insights applicable and useful for managers, scouts and financial planners.

(Li, Kampakis, & Treleaven, 2022) this study support machine learning as a modern alternative to conventional notational and video-based techniques for determining a football player's worth. Their research demonstrates that by examining large, multi-source datasets, models

such as Random Forest can identify intricate connections between player attributes, pay and market value. This shift to automated, data-driven assessment improves the consistency and transparency of player valuation procedures by enabling analysts to identify performance drivers that traditional methods frequently miss.

(Rivera, Santana, Burgueño, & Briseño., 2025) With a focus on forward players, this study develops models that illustrate how team dynamics, playing time and goal contributions affect market value. The findings highlight the value of segmenting valuation models rather than using a single strategy for all player roles by demonstrating that the use of metrics specific to a given position increases prediction accuracy.

In order to balance economic principles with the realities of transfer negotiations, (Follert & Gleißner, 2024) recommends for a decision-focused approach to football player valuation. In order to balance risk and reward, their model incorporates contractual, athletic and financial uncertainties. Their method offers both theoretical understanding and useful tools for enhancing club management and assessing player investments by coordinating investment choices with anticipated player performance, market circumstances and situational risks.

Football player market values are predicted using readily available metrics such as age, playing time, goals and assists in a study (Krishna & Chandran, 2021) that uses regression techniques. It draws attention to discrepancies between projected and actual valuations for well-known players like Neymar and Icardi, highlighting the challenges regression models have in accurately capturing the intricacies of player valuation. The study also emphasizes how the completeness of the dataset and the particular modeling technique employed can affect prediction accuracy.

According to research, age plays a significant role in player market value. Its nonlinear effect value increases early in a career, peaks mid-career, and falls with age-related performance declines is captured by taking into account both age and age squared. (Koen, Arem, Goes-Smit, & Söhl, 2025) stated how crucial the age squared variable is to their Estimated Transfer Value (ETV) model, which was constructed using professional football players biannual data. The model's predictive accuracy and applicability across a variety of player profiles are improved by adding age squared, which penalizes older players while capturing the development of younger players.

In sports analytics, estimating the market value of recently transferred football players without previous game data or recorded transfer fees is a major challenge. According to (Poli, Ravenel, & Besson, 2020), benchmarking and similarity-based analysis are essential when comprehensive quantitative data is lacking. By contrasting a new player with similar players those who share their age, position, physical characteristics, club history and geographic

background analysts can determine the player's market value. The premise behind this approach is that, especially when negotiating transfers for up-and-coming or less-established talent, the market values of established players in similar roles can be used as a guide.

The use of a combination of soft indicators to improve player valuation models is supported by recent advanced studies, such as those conducted by (Hill, Skinner, & Grosman, 2025) and (Franceschi, Brocard, Follert, & Gouguet, 2023). To create more comprehensive evaluations, these methods take into account factors like club prestige, market dynamics, contract specifics, physical characteristics and exposure to other countries. These contextual and qualitative cues assist in bridging the gap when performance data is scarce. Additionally, the models are made to be flexible, updating valuations in response to fresh competitive data.

In order to build precise models for forecasting football player market values, this dissertation makes use of real-world datasets from Fbref and Transfermarkt, which provide comprehensive, impartial and up-to-date statistics on player performance and market activity. A study from (Rijnsburger, 2024) compared player statistics, such as goals and minutes played, with data from the video game FIFA. It was discovered that real-world statistics more accurately depict shifting performance and market trends, even though FIFA data provides broad, standardized skill ratings. Prediction accuracy is increased by combining the two forms of data, but real-world data is still more trustworthy for accurate player valuation.

Therefore, real-world data from Fbref and Transfermarkt provides more accurate market value modeling, even though video game data aids in evaluating untested players.

In conclusion, this background review highlights the increasing difficulty and importance of precisely forecasting football player market values using data-driven methodologies. By using advanced machine learning techniques on extensive, multi-source datasets from platforms like Transfermarkt and FBref, this study addresses the drawbacks of conventional, subjective valuation methods. The project's emphasis on creating and evaluating different regression and ensemble models stems from the knowledge that contextual, demographic and performance factors need to be methodically included and examined. The stated goals data collection, exploratory analysis, model building, performance assessment and the creation of actionable insights reflect a practical approach meant to provide football teams and stakeholders with trustworthy, open and empirically supported valuation tools. The study adds to both research discussions and real-world decision-making in areas like contract negotiations, transfers, recruitment and strategic planning in the professional football landscape by pinpointing important factors that determine player value and improving prediction accuracy. Building on this foundation, the upcoming chapters describe the methodology, implementation and evaluation of the machine learning models created to accomplish these goals.



**Transfermarkt** is a reputable international platform that is well-known for estimating the market values of football players. It compiles information on player contracts, market trends, injury histories, transfer fees, and the financial stability of the club. Many people believe that the provided valuations accurately reflect current market conditions and provide an economic viewpoint that is frequently absent from metrics that only measure performance. The platform is an essential tool for financial analysis and valuation in football analytics since it also frequently releases projections for transfer fees and tracks the career paths of players. (<https://www.transfermarkt.com/schnellsuche/keinergebnis/schnellsuche?query=>)

The screenshot shows the Transfermarkt website interface. At the top, there is a search bar with the text "Enter your search term" and a "LIVE" indicator. Below the search bar, there are navigation tabs: DISCOVER, TRANSFERS & RUMOURS, MARKET VALUES, COMPETITIONS, STATISTICS, COMMUNITY, and GAMING. A secondary navigation bar shows filters for Country, Competition, Club, and Player. The main content area is divided into two sections: "NOTHING FOUND?" and "PLAYERS IN TREND".

**NOTHING FOUND?**

**THE FUNCTIONS OF THE QUICK SEARCH ARE SIMPLE**  
Competitions, clubs, persons (player, manager, agent, referee) can be entered in the search mask.

**IT IS NECESSARY TO KEEP IN MIND**  
When you search for a person, you need to type in the last name of the person first. If you did not find the desired result, we would recommend the detailed search: To the player detail search

**EXAMPLES FOR DETAILED SEARCH**  
When you type in "Messi,L" all Messi's will be listed, whose first names start with a "L".  
When you type in "Messi-PSG" all Messi's will be listed who play for "PSG". When you type in "Messi , L-PSG" all Messi's will be listed whose first names start with a "L" and who play for "PSG".

It is also possible to search for players from a club with specific shirt number. When you type in "Liverpool, 10" all players will be listed who play for Liverpool and wear the shirt number "10".

**NOTE**  
When you are entering only two letters (minimum), only exact matches will be listed.

**PLAYERS IN TREND**

|                                                                                                                                        |                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>Cristiano Ronaldo</b><br>market value: €12.00m | <br><b>Lionel Messi</b><br>market value: €18.00m       |
| <br><b>Mohamed Salah</b><br>market value: €50.00m    | <br><b>Ousmane Dembélé</b><br>market value: €90.00m   |
| <br><b>Kylian Mbappé</b><br>market value: €180.00m  | <br><b>Vinicius Junior</b><br>market value: €170.00m |
| <br><b>Raphinha</b><br>market value: €90.00m        | <br><b>Cole Palmer</b><br>market value: €120.00m     |
| <br><b>Lamine Yamal</b><br>market value: €200.00m   |                                                                                                                                           |

**QUICK LINKS**

- > Most valuable players in the world
- > Latest transfers
- > Latest rumours
- > Latest market value updates
- > FIFA World Ranking
- > Betting

**BE INVOLVED**

- > Mods & Data Scouts
- > Apply as Mod or Data Scout
- > 11 commandments
- > Found a mistake?

**SOCIAL MEDIA**

Icons for Instagram, Facebook, WhatsApp, X, and Telegram.

**CAREER**

- > Contact

**ABOUT US**

- > TM-Team
- > FAQ

Fig 2. Transfermarkt Data Source

### Following steps were followed for data preparation:

- **Data Scrapping:** To extract the Fbref data Python library Pandas was used in this project. Function `pd.read_html()` reads the HTML tables from the provided URL and the argument 'attrs' specifies the id of the table that needs to be scrapped.  
`pd.read_html('https://fbref.com/en/comps/Big5/stats/players/Big-5-European-Leagues-Stats', attrs={"id": "stats_standard"})[0]`
- Further the data was filtered to Eng Premier Football League and extracted into an excel file to be used further for training the Machine Learning models.
- To extract player market values from Transfermarkt, a customized Python script was developed utilizing the requests and BeautifulSoup libraries. A statistics DataFrame was used to loop through player names, create search URLs, and retrieve important information like player ID, name, position, age, nationality, club, and market value.
- **Data Merging:** The scraped data of Eng Premier League and Fbref data was merged to create a comprehensive dataset containing both performance statistics and market values.
- The target variable "Market Value" containing currency symbols and units ("€6.00m", "€500k"), was converted to numerical format representing the Market Values in Millions.

#### 6.1.2. Data Exploration and Feature Selection

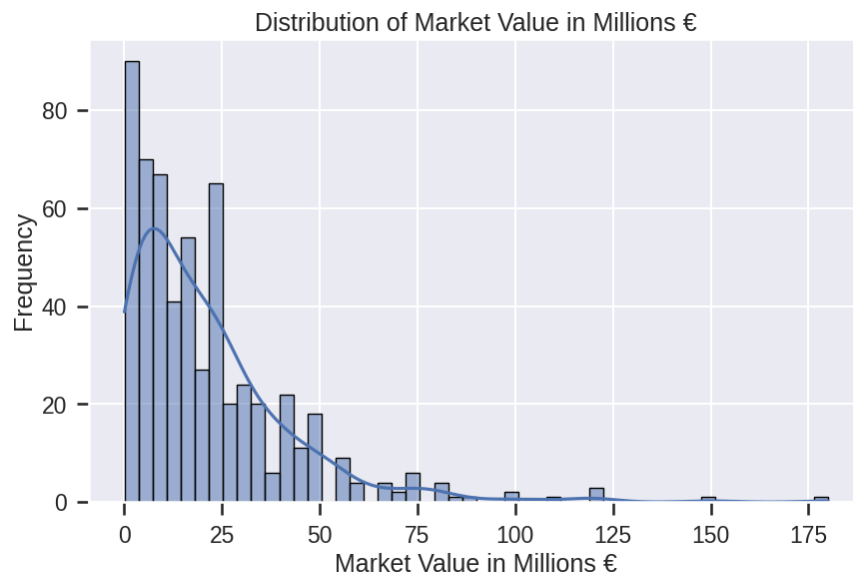
Moving towards the model development, a comprehensive exploratory data analysis (EDA) was implemented. This step provided important information about market value distributions, predictor variable statistical behavior, and possible associations with the desired result. In order to guarantee that only significant and well-structured variables were incorporated into the modeling process, the EDA was also essential in directing feature selection.

Using `.describe()` and `.info()`, the dataset was first analyzed using structural summaries and descriptive statistics. Data types, value ranges, central measures like mean and median, and dispersion indicators like standard deviation and quartiles were all identified with the aid of this procedure. Additionally, it revealed skewness in a number of variables, most notably the Market Value target, where a few top players displayed abnormally high valuations. The decision to use transformations later on, such as a logarithmic scale for market value, was supported by these findings.

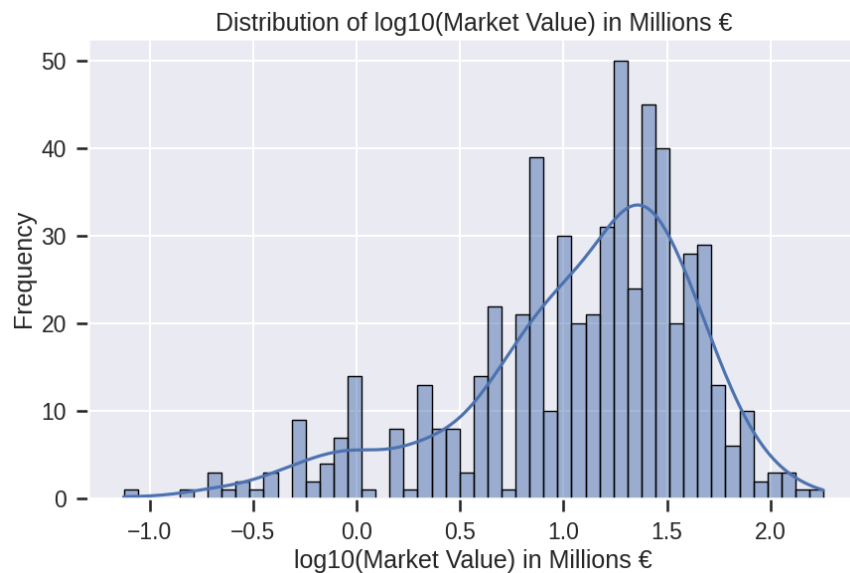
### Distribution Analysis:

For important numerical characteristics such as age, minutes played, goals and assists, histograms and Kernel Density Estimates (KDEs) were plotted in order to analyze data distributions. These visual aids highlighted outliers and deviations from the norm, such as abnormally high goal totals or exaggerated market values associated with top players.

Logarithmic conversion of Market Value was chosen as the primary dependent variable for regression modeling because of the target variable's distribution's notable right skewness.



*Fig 3. Distribution of Market Value without Log Transform*



*Fig 4. Distribution of Market Value with Log Transform*

## Correlation Analysis:

To investigate linear relationships between market value and numerical variables, correlation matrices were developed. A comprehensive view of inter-variable correlations was provided by a heatmap of all numerical features. In particular, distinct heatmaps were examined for Market Value and its log-transformed equivalent. The previous comparison demonstrated the skewness and outliers in the distribution chart, the same differences can be seen in the correlation heatmap when compared with raw market value and Log transformed Market Value.

The features (like npxG+AG, G+A, xAG, npxG) related to goals or assisted goals achieved by the player show a strong correlation with raw data of Market value.

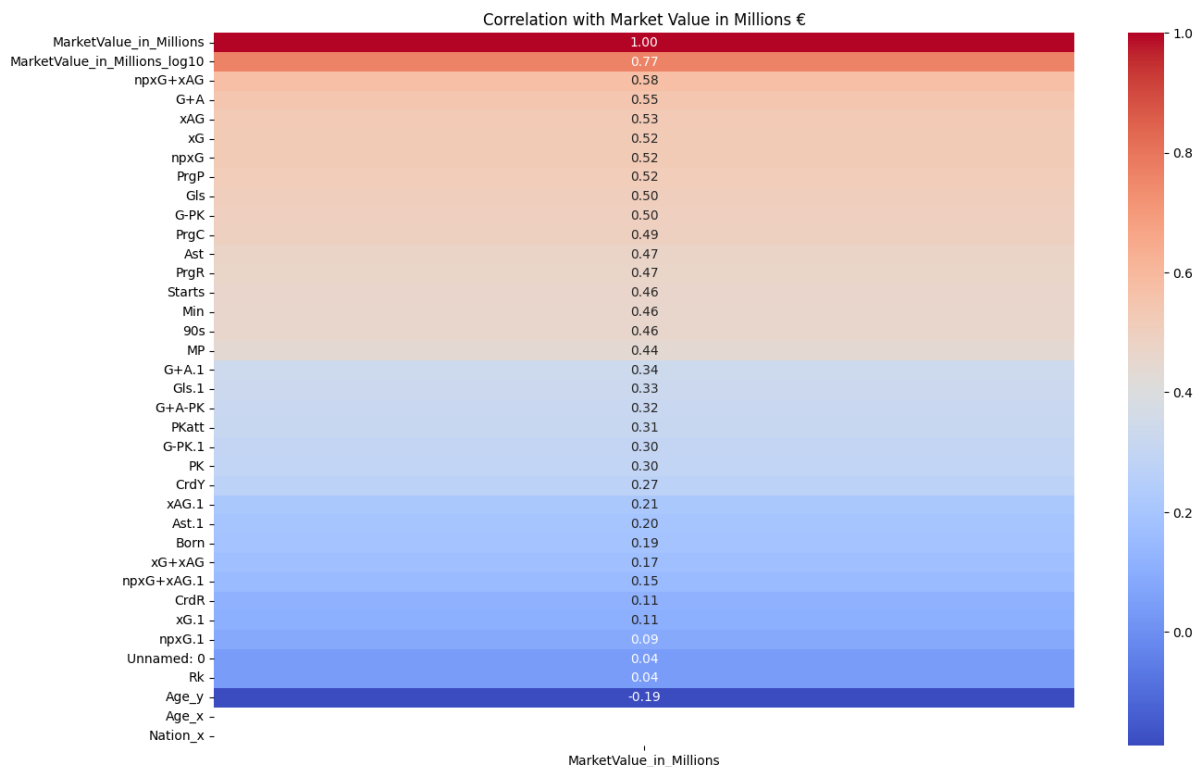


Fig 5. Correlation with Market Value without Log Transform

Similarly, the features representing (MP, min, 90s, starts, Prgp) the experience and performance of the player have a strong correlation with Logarithmic transformation of market value.

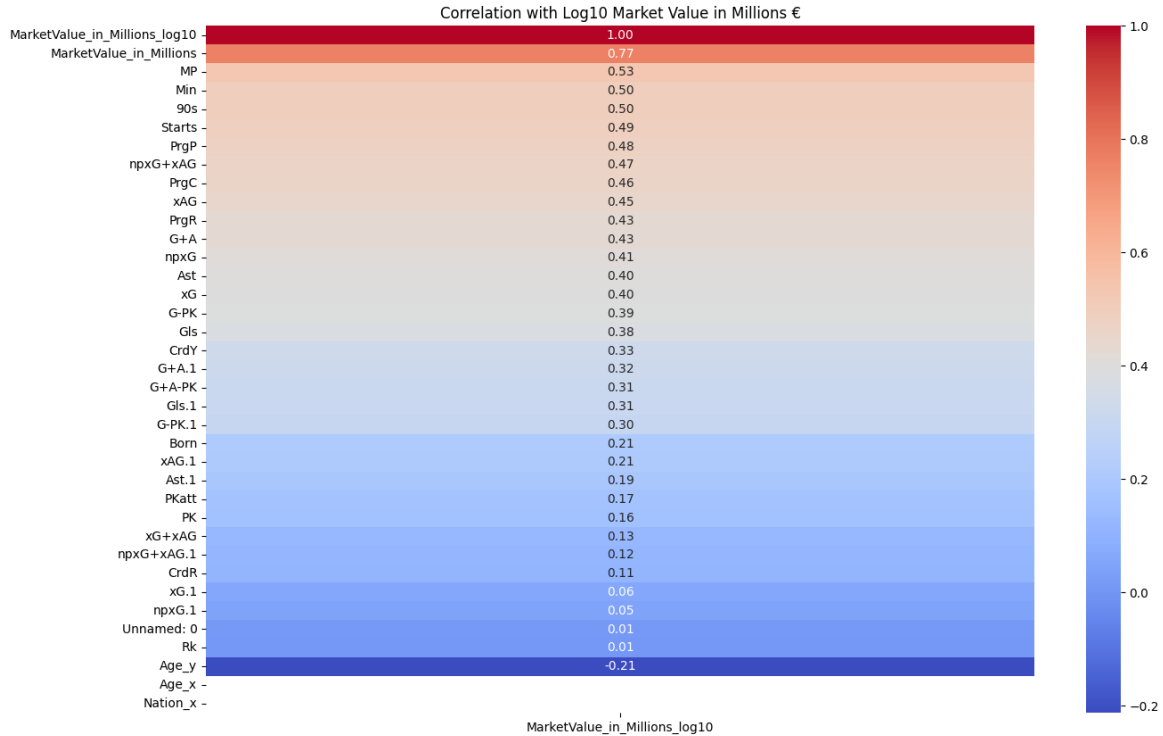


Fig 6. Correlation with Market Value with Log Transform

### Feature Engineering and Selection:

This stage is important to prepare the data for predictive modelling. The goal of this step is to identify the most important variables and format them according to the algorithms that were chosen. A set of numerical and categorical features was extracted from the data prepared previously, under the guidance of domain expertise and correlation analysis.

The numerical features that were kept included age (Age\_y) and performance metrics (e.g., MP, Starts, Gl's, Ast, xG, xAG, progressive stats). Nationality (Nation\_y), squad (Squad), and position (Pos) were among the chosen categorical variables.

To model the non-linear relationship between age and market value, a new variable called Age\_y\_squared was introduced. As this variable captures the normal rise, peak, and decline over the course of a player's career.

$$\text{'Age\_y\_squared'} = \text{'Age\_y'} ** 2$$

Regression analysis was performed on the finalized DataFrame, which included the target variables (MarketValue\_in\_Millions and MarketValue\_in\_Millions\_log10) as well as selected and engineered features. MarketValue\_in\_Millions\_log10 was selected as the main target because of its normalized distribution. After that, features (X) and target (Y) were kept apart for training and assessment.

### **6.1.3. Data Preprocessing**

After feature selection and data exploration, a structured preprocessing pipeline was used to get the dataset ready for model training. In predictive modeling, this step is essential, particularly when combining various feature types like numerical statistics, categorical attributes (like position or nationality), and contextual variables like club ranking. Regression and ensemble methods in particular can produce biased results, have trouble convergent, or perform poorly if they are not properly preprocessed (Li, Kampakis, & Treleaven, 2022). Encoding categorical variables, scaling numerical features and dividing the dataset for training and evaluation are the primary steps of the Python-based pipeline.

#### **Categorical Features:**

Features such as nationality and position were converted using one-hot encoding using scikit-learn's OneHotEncoder in order to prepare categorical variables for machine learning. Machine Learning algorithms are designed to work with numerical data and cannot directly handle text-based values. By guaranteeing compatibility between linear and tree-based models, this encoding technique supported a reliable and effective modeling pipeline.

#### **Numerical Features:**

Models that are sensitive to feature magnitude, like linear regression, Ridge, and Lasso, may be distorted by the wide scale variation of numerical variables like age, appearances, goals, assists, and minutes played. Each feature was normalized to a mean of zero and a standard deviation of one using scikit-learn's StandardScaler in order to account for this. During model training, this normalization guaranteed that each feature contributed equally and improved algorithmic convergence.

#### **Training and Test sets:**

The preprocessed dataset was divided into 80% of training and 20% of testing sets using `train_test_split` with a fixed random seed (42) in order to evaluate the generalizability of the model. The test set provided an objective assessment of predictive performance, while the training set assisted with parameter estimation and model fitting. By adhering to accepted machine learning techniques, this approach makes sure that the model's performance on unknown data is reflected in its efficacy rather than just fitting the training set (Kong, 2025).

### **6.1.4. Model Training**

#### **Linear Regression:**

Assuming a linear relationship between input features and the target variable, linear regression is the simplest technique for modeling player market value. This configuration

enables the model to estimate value through the additive contributions of each feature by representing a player's log-transformed market value ( $y_i$ ) as a weighted combination of predictors such as age, goals, and assists.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \epsilon_i$$

In this model,  $\beta_0$  represents the intercept,  $\beta_j$  represents the impact of each predictor, and  $\epsilon_i$  represents the error that cannot be explained. Because each coefficient accurately depicts the anticipated change in market value brought about by a one-unit shift in the corresponding feature, linear regression's main advantage is its interpretability (James, Witten, Hastie, Tibshirani, & Taylor, 2023). Its shortcomings are apparent, though, as football data frequently deviates from fundamental presumptions like independence and linearity, which lowers the model's dependability in intricate, real-world situations.

### **Ridge Regression:**

Ridge Regression is used to reduce multicollinearity, especially between predictors that show a high correlation, such as appearances and minutes played. By adding a penalty term that limits the squared magnitude of the coefficients, this method improves the standard linear regression framework, stabilizing estimates and enhancing model robustness.

$$\beta = (X^T X + \lambda I)^{-1} \{X^T\}_y$$

Large coefficient values are suppressed by the penalty term, which is controlled by  $\lambda$ , producing estimates that are more stable and trustworthy. Because of this, Ridge Regression is particularly helpful in sports analytics, where feature overlap is frequent because of comparable performance metrics. In contrast to Lasso, Ridge keeps all predictors, reducing their influence proportionately instead of eliminating them completely (Ghamrawi & Nat, 2024).

### **Lasso Regression**

Where Ridge Regression uniformly reduces the magnitude of all coefficients, Lasso Regression introduces an L1 regularization term that can shrink certain coefficients entirely to zero. This results in automatic feature selection, allowing the model to retain only the most influential predictors.

$$\min_{\beta} (\|y - X\beta\|_2^2 + \lambda \|\beta\|_1)$$

In addition to being a regression technique, Lasso is also a useful feature selection tool that automatically eliminates predictors with little use in determining market price. Lasso provides a simplified model that highlights the most important contributors in datasets with many

influencing factors, such as contextual variables, performance metrics, and demographics (Ghamrawi & Nat, 2024).

### Random Forest Regression

Apart from Linear models, Random forest provides a flexible and non-parametric method. Using a bootstrapped subset of the data and a randomly chosen set of features, this ensemble approach creates a large number of decision trees. The outputs from each individual tree are then averaged to determine the overall prediction.

Random Forests are particularly good at finding intricate relationships and non-linear patterns in the data, such as the combined effects of age and position on a player's market value. In contrast to more conventional regression techniques, they provide robust predictive performance and resistance to overfitting, but at the expense of decreased interpretability (Ghamrawi & Nat, 2024).

### Gradient Boosting Regression

Gradient Boosting builds trees sequentially, with each new tree being trained to fix the mistakes of the previous ones, in contrast to Random Forests, which combine predictions from numerous independent trees.

$$F_m(x) = F_{m-1}(x) + \nu h_m(x)$$

The model is updated iteratively, with  $F_m(x)$  representing the model at iteration  $m$ ,  $h_m(x)$  representing the weak learner, and  $\nu$  being the learning rate that establishes the relative contributions of each tree. Gradient Boosting can attain high accuracy and successfully capture the intricate, non-linear relationships present in football market valuation thanks to this methodical refinement. However, careful tuning is required because its performance is extremely sensitive to hyperparameters like learning rate and tree depth (Ghamrawi & Nat, 2024).

### Multilevel Model

The Mixed Linear Model framework was used in a multilevel modeling approach to take into consideration the hierarchical nature of the dataset, which shows that players are nested within clubs. This method allows for the simultaneous estimation of club-level influences (random effects) and player-level predictors (fixed effects).

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \beta_2 x_{2ij} + \dots + u_{0j} + \epsilon_{ij}$$

The log-transformed market value of player  $i$  in club  $j$  is indicated by  $y_{ij}$  in this formulation,  $x_{1ij}$ ... Individual-level predictors are represented by  $x_{kij}$ , a random intercept that captures unobserved variation at the club level is represented by  $u_{0j}$ , and the residual error is

represented by  $\epsilon_{ij}$ . According to the model, players who play for the same team enjoy common structural advantages like financial stability, notoriety, and prestige, which causes their valuations to be influenced by one another. The multilevel model provides deeper insights into the contextual drivers of market value by breaking down variance across both levels.

## **6.2. Tools and Technology**

Using the appropriate combination of tools and technologies to manage each step of the research from gathering and cleaning the data to analyzing and creating models was essential to the implementation of this dissertation. Because the project integrated concepts from machine learning, statistics, and sports economics, it required tools that could support both conventional and modern approaches.

Python's wide selection of open-source libraries made it the primary programming language. It provided the ability to work with sizable datasets, use various modeling techniques, and generate outcomes that were simple for others to duplicate. For tasks like web scraping, data organization, statistical testing, and visualization creation, additional tools were utilized. These resources worked together to create a robust and transparent research process from beginning to end.

### **6.2.1. Hardware**

A personal computer running Windows 11 with an Intel Core i7 processor, 16 GB of RAM, and a 1 TB SSD was used for all analyses. Enough processing power was available in this configuration to efficiently handle medium-sized datasets and train machine learning models. Despite the fact that certain algorithms, such as Gradient Boosting, are notoriously resource-intensive, the system managed them with ease and within acceptable time constraints. High-end servers and distributed computing were unnecessary because the local machine could handle the data and processing demands. This highlights the research approach's accessibility and reproducibility, demonstrating that sophisticated sports analytics can be carried out without the need for costly, high-performance computing infrastructure.

### **6.2.2. Google Colab**

During certain phases of data analysis and experimentation, Google Colaboratory (Colab) was utilized in the project. Colab is a free cloud-based platform that allows access to CPU and GPU resources and runs Python code in a Jupyter Notebook format. This was particularly helpful when training more complex models that would have taken longer to run on a personal computer alone, such as Random Forests, Gradient Boosting, and XGBoost.

The dissertation achieved accessibility and efficiency by integrating cloud-based tools with local computing. This hybrid configuration demonstrates how contemporary sports analytics can improve conventional research workflows by utilizing free cloud technologies.

**Colab also provided with a number of significant advantages as follows:**

**Reproducibility:** Without requiring local installation, code, data, and outcomes could be shared and executed consistently across various devices.

**Scalability:** The training of intricate ensemble models was accelerated by having access to GPU resources.

**Collaboration:** Work could be directly saved and versioned in Google Drive thanks to the cloud-based setup, which made managing it simpler.

### **6.2.3. Frameworks and libraries**

Python and its extensive library of open-source libraries were used to build the analytical framework for this dissertation. Every step of the research process was aided by these tools, including data collection and cleaning, statistical analysis, machine learning model construction, and visualization. Every library was selected on the basis of its dependability, extensive usage within the data science community, and capacity to satisfy the particular requirements of football player market value prediction.

**Pandas:** A key component of handling and processing the data was the pandas library. Combining player performance statistics from FBref with market value data from Transfermarkt was made simple by its ability to handle tables and spreadsheets with ease. Pandas was also used to organize the features required to create predictive models and to correct data problems, such as converting textual market values into correct numbers.

**NumPy:** The project's numerical foundation was NumPy, which provides quick and effective array and linear algebra operations. It was essential for tasks like matrix manipulation and feature scaling, particularly in regression modeling.

**Scikit-learn:** The primary library for implementing machine learning was scikit-learn, also known as sklearn. It supported a number of crucial features:

- **Preprocessing:** includes using OneHotEncoder to handle categorical data, StandardScaler to standardize numerical features, and OneHotEncoder to fill in missing values.
- **Model building:** includes training both more complex ensemble techniques like Random Forest and Gradient Boosting as well as fundamental regression models like Linear, Ridge, and Lasso.

- **Pipelines:** Preprocessing and model training were automated to streamline the entire workflow and stop data leaks.
- **Evaluation:** Assessing model performance with metrics such as R2, RMSE, and Mean Absolute Error (MAE).

When combined, these libraries offered a strong and trustworthy foundation for creating and evaluating prediction models in football market value analysis.

**Statsmodel:** The Multilevel (Mixed Effects) Model was constructed using Statsmodels, which necessitated precisely defining both fixed and random effects. Statsmodels, as opposed to scikit-learn, provides statistical inference tools, including the ability to interpret coefficients, compute confidence intervals, and conduct hypothesis testing. This makes it particularly helpful for preserving the dissertation's methodological depth.

**BeautifulSoup:** For web scraping tasks, BeautifulSoup and Requests were combined. While BeautifulSoup parsed the HTML content to extract structured market value data from Transfermarkt, Requests managed the web page downloads. Effective and scalable data collection was made possible by this configuration, which was also adaptable to various webpage formats.

**Matplotlib, Seaborn:** For presenting important findings and visualizing data during exploration, Matplotlib and Seaborn were indispensable. Seaborn expanded on Matplotlib's comprehensive plot design control by offering simpler methods for producing statistical charts such as distribution plots, boxplots, and heatmaps. These tools were particularly helpful in identifying multicollinearity among features, identifying skewed patterns in the target variable, and demonstrating the relationship between player positions and market value trends.

#### 6.2.4. Version Management

The code, data, and configuration files for the project are managed in a Git repository. This version control system makes it possible to keep track of modifications, roll back to earlier iterations, and, if necessary, facilitate collaboration. The repository is housed on GitHub, which offers a dependable version control platform.

Git: [https://github.com/Pranav216/Dissertation\\_19247207.git](https://github.com/Pranav216/Dissertation_19247207.git)

Google Drive has been used as an additional tool to store project documents, including presentations, progress reports, and literature reviews, in addition to Git. By adding an additional layer of redundancy, this method protects against possible data loss from machine corruption or unintentional deletion within the Git repository.

Google Drive:

[https://drive.google.com/drive/folders/1gB9hyT47RLJ5hVheaHOngluTcVtWg8h9?usp=drive\\_link](https://drive.google.com/drive/folders/1gB9hyT47RLJ5hVheaHOngluTcVtWg8h9?usp=drive_link)

## 7. Results

### 7.1. Evaluation Metrics

Five models were evaluated in this implementation: Random Forest Regressor, Lasso Regression, Ridge Regression, Linear Regression, and Gradient Boosting Regressor. Standard regression metrics were used to assess each model's performance, and scatter plots comparing expected and actual market values were used to illustrate the accuracy and suitability of each strategy.

Models were evaluated on the test set using several key performance metrics:

**MAE:** The average amount of the discrepancies between expected and actual log-transformed market values is measured by the Mean Absolute Error, or MAE. The model is more accurate when the values are smaller.

**MSE:** The average of the squared discrepancies between expected and actual values is determined by the Mean Squared Error (MSE). Better performance is indicated by lower scores because it penalizes bigger errors more severely than MAE.

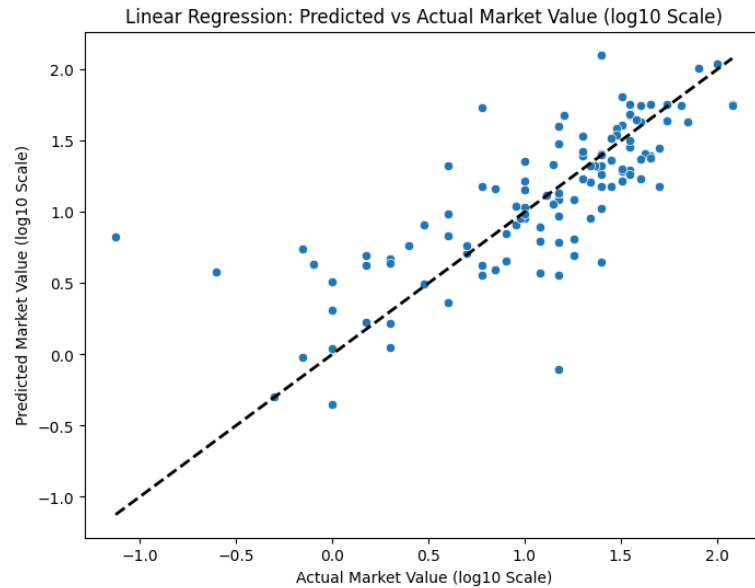
**RMSE:** By taking the square root of the MSE, the Root Mean Squared Error (RMSE) provides an estimate of the typical error size on the same scale as the target variable. Once more, greater accuracy is reflected in lower values.

**R<sup>2</sup>:** R-squared Indicates the extent to which the model's predictors can account for the variation in market value. While values near 0—or even negative—indicate poor predictive power, values closer to 1 indicate a better fit.

### 7.2. Model Results

#### 7.2.1. Linear Regression

Linear Regression Captured about 57% of the variance in the log-transformed market value. Although not very high, it shows that a respectable amount of the relationship is captured by the linear model.



*Fig 7. Linear Regression Scatter plot (Actual vs Predicted Market Value)*

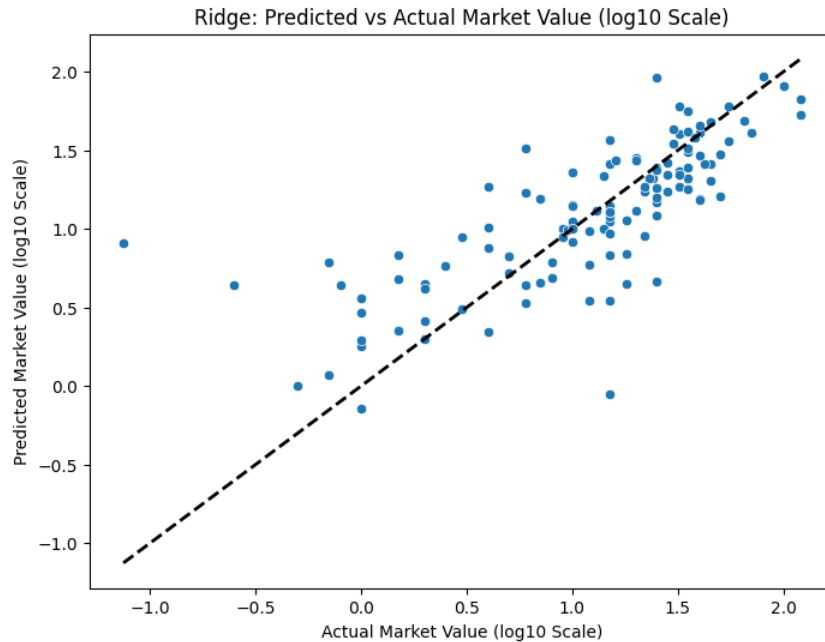
```
Evaluation results for Linear Regression:  
Mean Absolute Error (MAE): 0.26  
Mean Squared Error (MSE): 0.15  
Root Mean Squared Error (RMSE): 0.39  
R-squared (R2): 0.57  
-----
```

*Fig 8. Linear Regression Evaluation results.*

Based on the above evaluation results for Linear Regression, the metrics reflect moderate accuracy in estimating log transformed Market Values.

### **7.2.2. Ridge Regression**

Ridge regression had a somewhat higher R-squared (0.58) than standard linear regression, but otherwise performed very similarly. By reducing coefficients toward zero without precisely setting them to zero, Ridge's L2 regularization helps avoid overfitting.



*Fig 9. Ridge Regression Scatter Plot (Actual vs Predicted Market Value)*

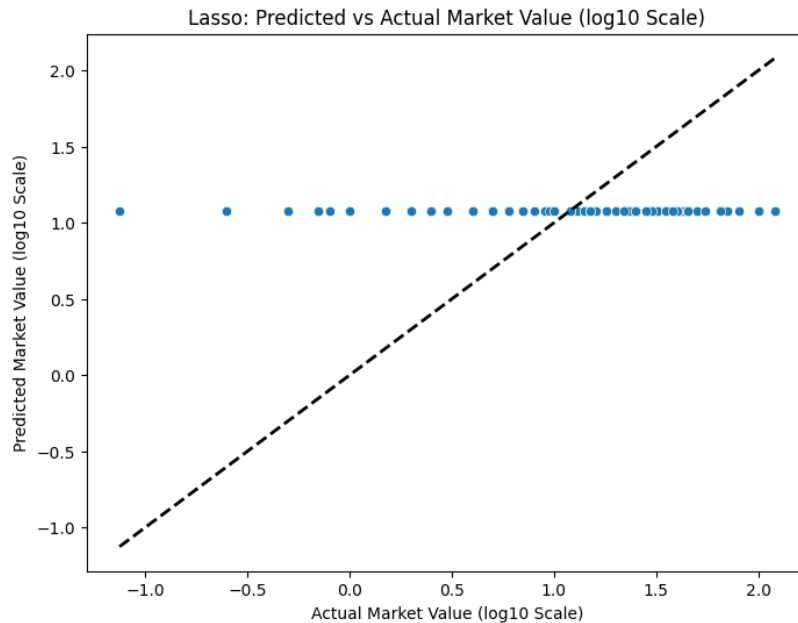
```
Evaluation results for Ridge:  
Mean Absolute Error (MAE): 0.26  
Mean Squared Error (MSE): 0.15  
Root Mean Squared Error (RMSE): 0.38  
R-squared (R2): 0.58
```

*Fig 10. Ridge Regression Evaluation results.*

Given the small performance difference, it is possible that multicollinearity is not a significant problem or that Ridge's alpha (regularization strength) was not having a significant effect on the model's fit when compared to basic linear regression.

### **7.2.3. Lasso Regression**

With a very low (near zero or negative) R-squared and noticeably higher error metrics than Ridge and Linear Regression, the Lasso model did not perform well. Lasso employs L1 regularization, which essentially performs feature selection by shrinking some feature coefficients to zero.



*Fig 11. Lasso Regression Scatter Plot (Actual vs Predicted Market Value)*

```

Evaluation results for Lasso:
Mean Absolute Error (MAE): 0.46
Mean Squared Error (MSE): 0.35
Root Mean Squared Error (RMSE): 0.59
R-squared (R2): -0.00

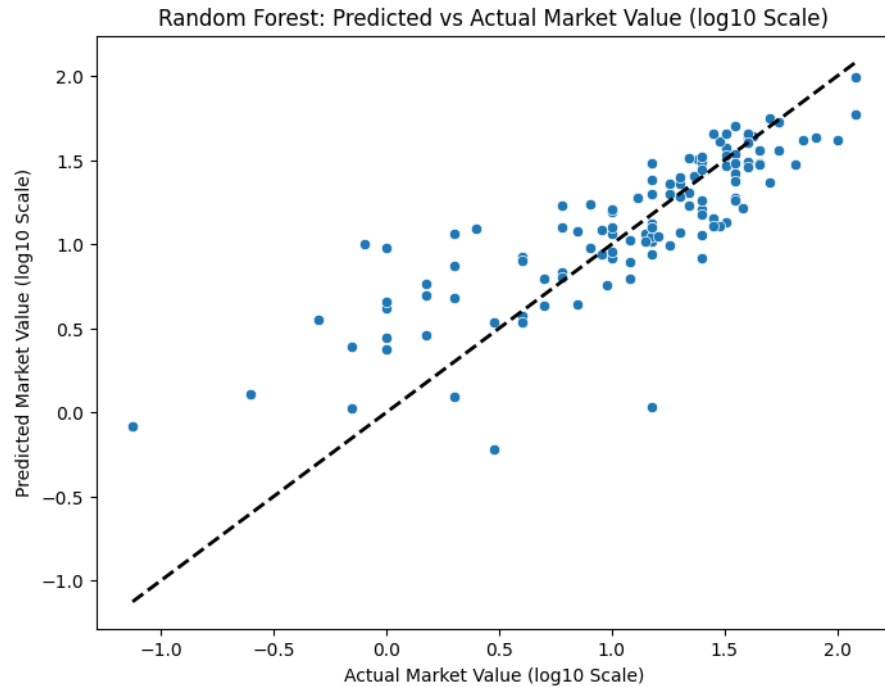
```

*Fig 12. Lasso Regression Evaluation results*

According to the results, the Lasso model, possibly using the default regularization strength, was too harsh and eliminated or unduly penalized features that were crucial for prediction, producing a model that explains essentially none of the variance in the target variable.

#### **7.2.4. Random Forest Regression**

The Random Forest model outperformed the linear models by a significant margin. With a higher R-squared of 0.67 and lower error metrics, it accounts for 67% of the variance in the log-transformed market value.



*Fig 13. Random Forest Regression Scatter Plot (Actual vs Predicted Market Value)*

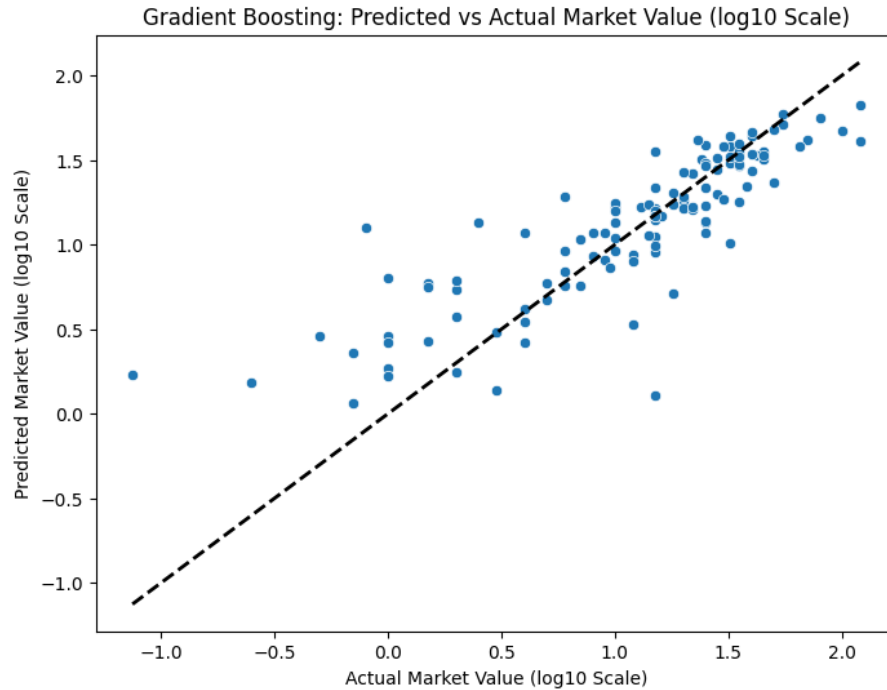
```
Evaluation results for Random Forest:  
Mean Absolute Error (MAE): 0.24  
Mean Squared Error (MSE): 0.11  
Root Mean Squared Error (RMSE): 0.34  
R-squared (R2): 0.67
```

*Fig 14. Random Forest Regression Evaluation results*

Based on the visual and readings, random Forest performs better on this dataset because, as an ensemble of decision trees, it can capture non-linear relationships and interactions between features that linear models are unable to.

#### **7.2.5. Gradient Boosting Regression**

Out of all the models that were tested, the Gradient Boosting model performed the best. It has the highest R-squared (0.70) and the lowest MAE, MSE, and RMSE.



*Fig 15. Gradient Boosting Regression Scatter Plot (Actual vs Predicted Market Value)*

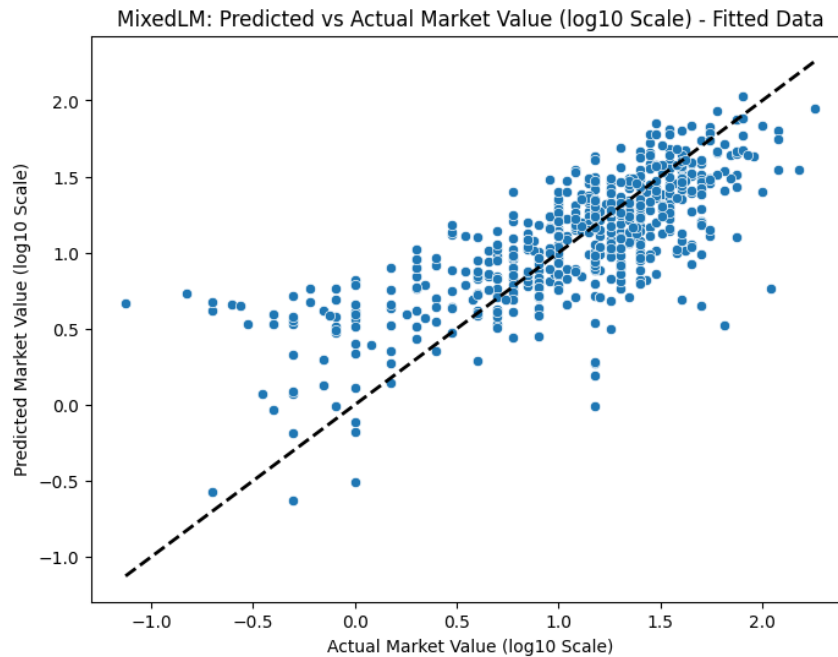
```
Evaluation results for Gradient Boosting:  
Mean Absolute Error (MAE): 0.21  
Mean Squared Error (MSE): 0.10  
Root Mean Squared Error (RMSE): 0.32  
R-squared (R2): 0.70
```

*Fig 16. Gradient Boosting Regression Evaluation results*

Gradient Boosting constructs trees in a stepwise manner, fixing the mistakes of the trees that came before it. Predictions made through this iterative process are frequently very accurate. Based on the given features, the model's R-squared of 0.70 indicates that it can account for 70% of the variation in the log-transformed market value.

#### **7.2.6. Multilevel Regression**

This project also attempts to implement and test a Multilevel Regression model using Mixed Linear Model framework and the results were quite similar to the previous linear regression and Ridge regression.



*Fig 17. Multilevel Regression Scatter Plot (Actual vs Predicted Market Value)*

```

Evaluation results for Multilevel:
Mean Absolute Error (MAE): 0.27
Mean Squared Error (MSE): 0.14
Root Mean Squared Error (RMSE): 0.37
R-squared (R2): 0.56
  
```

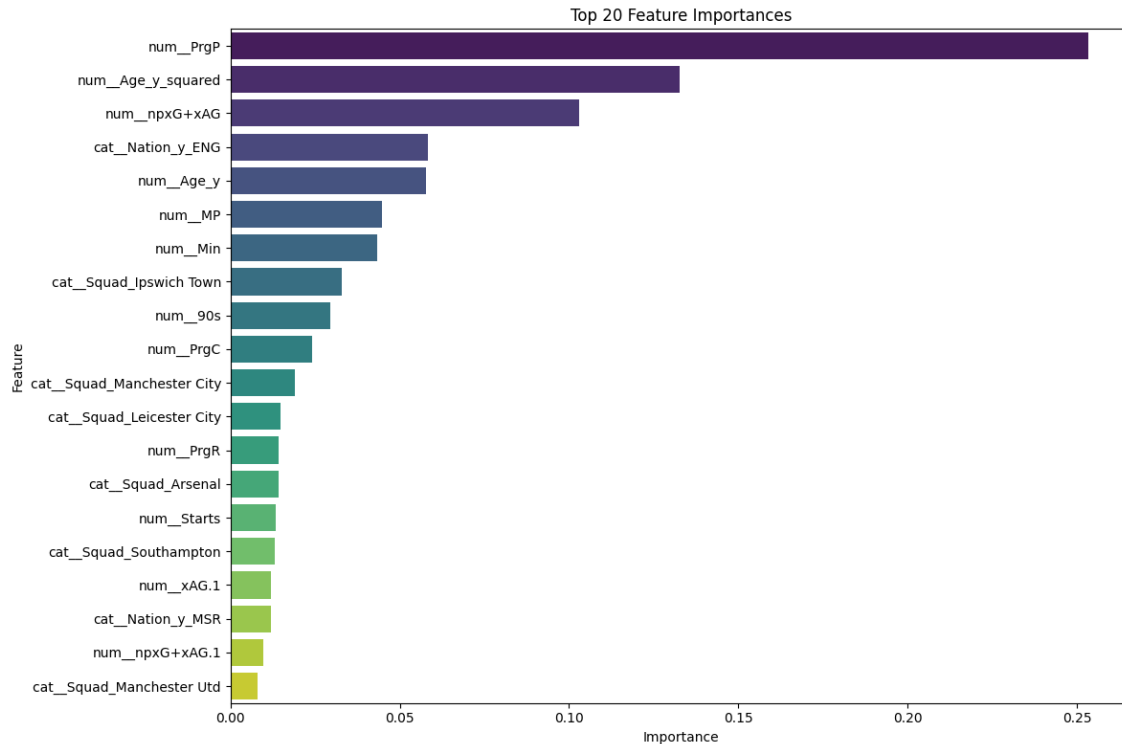
*Fig 18. Multilevel Regression Evaluation results*

Multilevel Regression recognizes the nested structure of the data to predict the market value. Here Players market value is influenced by their individual characteristics i.e. Level 1- features and also their club characteristics i.e. Level 2.

Based on this process the results of predictions are quite similar to a Linear Regression model giving an RMSE value of 37% and  $R^2$  value of 56%.

### 7.3. Discussion of Results

Based on the best performing model i.e. Gradient Boosting, the implementation further calculates the importance score, to measure how much each feature contributes in regards to reducing the model's overall error.



*Fig 19. Top 20 features Importance (Gradient Boosting)*

Following Features had the most influence to reducing Gradient Boosting models error: **PrgP (Progressive Passes)**: Progressive Passes highlights player's ability to break defensive lines and push on the ball into attacking areas. Even if the players don't progress with many goals or assists, clubs can identify the players who can excel at ball control and progression.

Talent scouts can identify and prioritize the leaders who progress possession, as this ability is not only rare but also central to team tactical advantages.

From investment perspective, executive can rely to allocate greater financial resources towards players with such stats to enhance team control and offence structure.

**Age<sup>2</sup>**: This attribute explains the peak market value typically occurs at a certain age, and later discontinued due to decline in resale value.

Clubs can optimize buying/selling cycles, negotiate deals around peak age windows to ensure prime years are secured and late career contracts avoided.

**npxG + xAG (Non-Penalty Expected Goals plus Assisted Goals)**: These attribute captures both shot quality and creative output. These profiles identify players with strong potential to thrive in more competitive environment.

Club analysts can assess the investments in attacking personnel by reliable and repeatable performance rather than short term statistical spikes.

**Nation\_ENG (Nationality England):** This explains that English players tend to command higher market values (maybe due to Premier league data and it's local fan demand).

With the recent studies from (Koen, Arem, Goes-Smit, & Söhl, 2025) and (Kong, 2025) the results are consistent and examine how well tree-based ensemble methods, in particular Gradient Boosting, assess player market value. Especially when working with complex feature interactions and non-linear correlations, which are typical in football data, these approaches show excellent prediction accuracy and provide deeper insights into how individual factors contribute to outcomes.

because of in large part to their inability to capture the intricate, non-linear patterns and layered variable connections that define real-world football market data, conventional methods such as linear and lasso regression performed poorly in terms of prediction. Due to these constraints, it was challenging for these models to accurately capture the processes affecting player valuation.

Although the Gradient boosting was the best performing model, Multilevel model on the other hand captures the data in a hierarchical or nested structure.

|     | Feature    | Coefficient | P-value  |
|-----|------------|-------------|----------|
| 97  | G_PK_1     | -20.868037  | 0.451745 |
| 98  | G_A_PK     | 19.270886   | 0.504387 |
| 94  | Gls_1      | 15.933495   | 0.560906 |
| 96  | G_A_1      | -14.073331  | 0.621926 |
| 103 | npxG_xAG_1 | 13.765134   | 0.142242 |
| ... | ...        | ...         | ...      |
| 77  | Min        | -0.001053   | 0.858848 |
| 91  | PrgC       | 0.000843    | 0.537403 |
| 92  | PrgP       | 0.000816    | 0.201024 |
| 93  | PrgR       | -0.000725   | 0.353167 |
| 85  | CrdY       | -0.000218   | 0.978822 |

Fig 20. Fixed Effects Sorted by Absolute Coefficient

|  | Feature              | Coefficient | P-value      |
|--|----------------------|-------------|--------------|
|  | Age_y_squared        | -0.006109   | 5.015638e-24 |
|  | Age_y                | 0.286098    | 9.140762e-20 |
|  | Intercept            | -2.290232   | 1.461225e-05 |
|  | Nation_y_MSR[T.True] | -1.506312   | 1.159584e-03 |
|  | MP                   | 0.010759    | 1.268419e-02 |
|  | ...                  | ...         | ...          |
|  | Gls                  | 0.758414    | 9.999964e-01 |
|  | Ast                  | 1.330353    | NaN          |
|  | G_A                  | -1.323613   | NaN          |
|  | G_PK                 | 0.579053    | NaN          |
|  | PK                   | 0.500000    | NaN          |

Fig 21. Fixed Effects Sorted by P-value

Like a typical regression model, Fixed Effects represent the average associations between the log10 market value across all clubs and the individual player attributes. Here the coefficient shows the estimated change in Target value (i.e. Log10 of Market Value), and P-value

calculates the statistical significance of the coefficient. A coefficient represents how much a feature is associated with the target variable. Whereas, P-value tells you how confident you can be that the association is not just random noise. The prediction of the Multilevel model can help predict players market value in more rigorous way to understand the club influence towards individual player statistics. But having lower prediction accuracy than a tree based model, it makes the model less reliable.

This work demonstrates the effectiveness of ensemble-based machine learning methods for football teams, especially Gradient Boosting in conjunction with thorough feature significance analysis. In a very complex and dynamic context, these techniques offer a clear, evidence-based framework for assessing player market value, improving recruitment tactics, and reducing transfer-related risks. The results complement and broaden previous scholarly work that continuously emphasizes the benefits of sophisticated, interpretable models in football analytics.

#### **7.4. Issues Encountered**

The effort to use automated web scraping techniques to collect market value data from Transfermarkt was one of the main challenges encountered throughout this project. Despite several attempts, an HTTP 403 Forbidden error that surfaced after just a few queries repeatedly stopped the procedure. This answer demonstrated the site's strict protections against automated data collecting and successfully prevented additional access (TransferMarkt). To tackle this issue, player statistics data from Fbref is reduced and filtered to English Premier League and Transfermarkt data is scrapped by iterating the player names in a loop, further saving the dataframe in excel.

Another issue this project faced was that raw market value distribution was highly skewed, it was clearly visible and known that a small number of players had high value compared to average which resulted in outliers. To address this issue, logarithmic transformation was applied to compress large values and spread the smaller ones, this lead to approximating a normal distribution, reducing the impact of outliers and stabilizing variance.

#### **7.5. Constraints and Limitations:**

There were several constrains and limitations noted in this project.

- Data accessibility can be a concern since scraping data from websites like Transfermarkt was restricted by 403 errors, which made it difficult to extract data completely automatically. Instead, personal intervention and rigorous adherence to conditions of use were necessary.

- Because predictions could be less accurate for player groupings that differ from the training data in terms of region or league, the model's generalizability can also be constrained.
- The quality and scope of the information utilized were impacted by the discrepancies and incompleteness of the publicly accessible data from sources such as Fbref and Transfermarkt, making it challenging to capture all pertinent player characteristics and contextual elements.

These constraints highlight the need for improved data gathering tactics, the inclusion of more detailed and contextual characteristics, and ongoing validation to increase model resilience and real-world application in future research.

## **7.6. Originality**

This dissertation provides novel integration of multiple real world data sources paired with powerful ensemble learning models to predict football player market prices. This method prioritizes openness and interpretability, providing a more rigorous framework than earlier studies that relied on crowd-sourced values, video game measurements, or traditional regression approaches.

The research finds that Gradient Boosting is the best accurate approach for estimating player market value after a systematic examination of linear, regularized, and tree-based modelling methodologies. In addition to its great performance, the model produces unambiguous rankings of feature significance, which provides useful interpretability. The findings are particularly beneficial for decision-makers in football operations, from scouting departments to executive leadership, due to their emphasis on predictive precision and openness.

In conclusion, this dissertation stands out for its innovative use of complex statistical tools, modern performance measurements, and a comprehensive grasp of football market dynamics. These aspects combine to provide a valuation framework that outperforms traditional methodologies in terms of clarity, empirical depth, and interpretability. In addition to promoting scholarly discussion, the method offers professionals working in player evaluation and transfer strategy useful information.

## **8. Professional Issues**

### **8.1. Project Management**

#### **Schedule**

This project is structured into multiple phases, each designed with specific objectives, tasks and timelines. The following section breakdowns a detailed schedule of planned activities, duration, weekly time commitments and start- end dates.

| Activity                               | Duration in weeks | Hrs. per week (approx.) | Start Date   | End Date     |
|----------------------------------------|-------------------|-------------------------|--------------|--------------|
| Literature Review                      | 12                | 20 - 30                 | 6/May/2025   | 31/July/2025 |
| Data Collection/ Scrapping             | 4                 | 35                      | 9/June/2025  | 6/July/2025  |
| Data Exploration and Feature selection | 3                 | 30                      | 7/July/2025  | 27/July/2025 |
| Data Preprocessing                     | 2                 | 40                      | 28/July/2025 | 10/Aug/2025  |
| Model Training                         | 2                 | 40                      | 11/Aug/2025  | 24/Aug/2025  |
| Model Evaluation/ Visualization        | 4                 | 30 - 40                 | 11/Aug/2025  | 7/Sept/2025  |
| Presentation and Dissertation          | 2-3               | 45                      | 8/Sept/2025  | 26/Sept/2025 |

## Data Management

For this project a Google drive folder and Github repository was created. Drive and Repository is organized as follows

- **Dataset** includes scrapped data from the sources Fbref and Transfermarkt website.
- **Code files** for Market Value Analysis and Data Scrapping is present.
- For **Version Control** GitHub was used, and primary coding was done on Google Colab.

## Deliverables

**ML Models:** Linear Regression, Ridge Regression, Lasso Regression, Random Forest regression, Gradient Boosting regression and Multilevel Regression are trained and evaluated models as a key deliverable.

**Final Thesis:** A Dissertation/ Thesis compiling the study's research framework, methodology, issues encountered and final conclusions.

## 8.2. Risk Analysis

Based on the issues and challenges faced in this project, there are various risks to be considered as follows:

**Data Availability:** Using public website like Transfermarkt and Fbref comes with risks, these sites may block automated data collection, change their access rules or limit the available

data. These things can result in missing information, which can lead making the model less accurate and difficult to rely on.

During the project's implementation, the data scrapping from transfermarkt occasionally had 403 error which is a common issue as scrapping data from transfermarkt is difficult due their Terms of Service. To address this, the best practice is to save the extracted data in the excel to avoid errors in every run, later proceed to test and train the models.

**Model Performance:** As seen in the above sections, skewness and multicollinearity can be common in such data, if not handled properly this can result in overfitting and distorted coefficient estimates. Addressing these issues is essential, ensuring the model robustness.

Addressing this issue of skewness in the target value in this project, a logarithmic transformation was practiced helping in providing normal distribution for the target value, giving the model stable market values for better performance.

**Legal and Ethical:** Collecting or Scrapping data by automating the process can violate their terms of service, which may lead to legal issues and can also lead to 403 errors (restricted access). This highlights the importance of respecting data ownership and ensuring ethical compliance throughout the research.

### 8.3. Future Risks

Although this project has progressed with data scrapping, training prediction models and evaluating those models, there may be some possible risks involved if used with other or updated data in the future.

**Data Quality and Completeness:** gaps and irregularities in data structure can weaken the reliability of prediction models. These leads to noisy and uncertain data, giving larger error margins and weak valuations.

**Generalizability:** A model may not predict target values very well for players with no data present, making it less useful in fresh football markets around the world.

Staying within the legal boundaries, using adaptable data strategy, preprocessing and validation procedures can be best practice to reduce these risks. And documenting these risks and limitations promotes transparency for decision makers to understand where the model predictions are most reliable and where it may fall short.

### 8.4. Legal Issues

One essential requirement for data-driven research is guaranteeing legal compliance, especially when obtaining data from outside sources like Transfermarkt. The UK's (BCS, 2022) Code of Conduct and the UK Research Integrity Office (UKRIO, 2025) Code of Practice both

emphasize the significance of abiding by pertinent regulations, such as those pertaining to intellectual property rights and data protection, like the GDPR. This project complies with these standards.

Transfermarkt's limitations on automated scraping created difficulties throughout the data collecting stage, leading to HTTP 403 failures. This study upheld its ethical integrity and complied with UK regulatory requirements by adhering to these restrictions and preventing unwanted access. Ignoring such responsibilities might result in legal consequences and limited access to data, endangering the study's validity and continuation.

### **8.5. Ethical Issues**

Design and implementation are ethically guided by transparency, diversity, and responsible data usage. According to (UK Data Service) the UK Economic and Social Research Council's (ESRC) Framework for Research Ethics, research procedures must minimize damage, uphold participant rights, and be honest and accountable.

This study followed ethical best practices by avoiding the abuse of potentially biased or incomplete data, revealing data limitations, and making sure interpretations are clear and realistically contextualized, even if it employed publicly available secondary data. The (Government Digital Service, 2020) UK Data Ethics Framework's ethical guidelines also require that analyses show public benefit and fairness, ensuring that they don't reinforce prejudice or injure underrepresented groups.

### **8.6. Environmental Issues:**

Prioritizing environmental factors is just as important as considering other professional issues, the (Government Digital Service, 2020) promotes resource responsibility. By increasing computational efficiency during model training and lowering duplicate processing operations, this research exhibits environmental awareness and lowers the energy needs generally associated with cloud-based computing. Such actions align with the UK's larger effort to incorporate sustainability considerations into digital research ethics.

### **8.7. Social Issues**

In a data-based study, social responsibility is acknowledging and addressing the broader impacts for people and communities. In order to prevent algorithmic modeling from enhancing already existing societal views, UK ethical frameworks including guidelines from the (Royal Statistical Society , 2019) place a strong emphasis on justice, inclusion, and openness.

In football analytics, this means creating player value models that promote fair assessment, enhance professional opinion, and prevent players from being disadvantaged by distorted or

biased data sources. Building trust and ensuring morally appropriate implementation need open and honest communication with stakeholders about the model's constraints and scope.

The project's spot inside reputable UK professional frameworks demonstrate a deep dedication to social responsibility, ethical behavior, environmental sustainability, and legal compliance. For data science research to be credible and have the potential to have a significant, responsible influence, these pillars are necessary.

## 9. Conclusion

The study presents a data-driven approach to evaluating football player market values using machine learning techniques. The study found that Gradient Boosting Regression outperformed linear, regularized, and ensemble models, with a  $R^2$  of around 0.70. This finding emphasizes the need of incorporating non-linear correlations and interaction effects to improve prediction performance and model interpretability.

The evaluation of feature importance found that progressive passes, age<sup>2</sup> (a non-linear age effect), non-penalty anticipated goals plus assisted goals (npxG + xAG), English nationality (most likely due to English Premier League data), and minutes played are important drivers of market value. These findings suggest that advanced performance indicators are more predictive than traditional metrics, age-related patterns reflect career paths, nationality may indicate structural or commercial factors, and consistent playing time serves as an indicator for managerial trust and player reliability.

This study helps to understand how prediction models perform in comparison to their nature. Showing that tree-based models capture complex interactions more effectively than linear models, mostly due to its iterative approach. This shows that effectiveness of predictive results is highly related to the anatomy of the modelling technique.

Even so, the model has limits. It does not yet include injury history, or qualitative characteristics like leadership and reputation. Future study might fill these gaps by including more detailed contract and injury data.

In conclusion, this research provides a rigorous and usable technique for player valuation, which contributes to both academic discussion and professional practice in sports analytics and football economics. It provides a framework for more transparent, efficient, and evidence-based decision-making in the ever-changing landscape of the football transfer market.

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